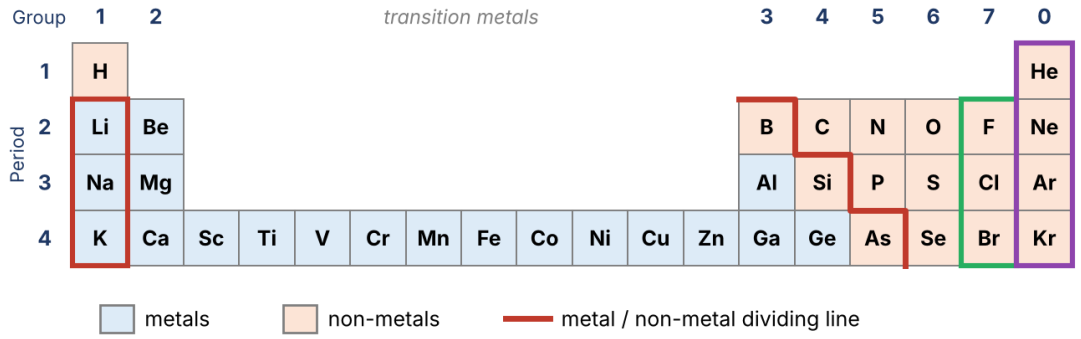
Organisation of the Periodic Table

AQA GCSE Combined Science: Trilogy · Chemistry Topic 1 · Revision activity 5 · 18 marks · about 25 minutes

Read the key information (5 min), answer the **quick recall** questions without looking back (5 min), then answer the **exam questions** (15 min). Mark your work with the mark scheme on the last page.

Key information

- Elements are arranged in order of **atomic number**.
- The vertical columns are called **groups**. Elements in the same group have the **same number of electrons in their outer shell**, so they have **similar chemical properties**.
- The **group number** = the number of electrons in the outer shell (except group 0, which has a full outer shell).
- The horizontal rows are called **periods**. The **period number** = the number of electron shells.
- **Metals** are on the left and towards the bottom. **Non-metals** are towards the **right and top**. Most elements are metals.
- Metals react to form **positive ions**. Non-metals do not form positive ions.



Typical properties of metals and non-metals

Metals	Non-metals
shiny	dull
good conductors of electricity and heat	poor conductors (insulators)
high melting and boiling points (most)	low melting and boiling points (most)
malleable (can be hammered into shape)	brittle when solid
form positive ions	do not form positive ions

Worked example

Sulfur has the electronic structure **2,8,6**. 3 shells → **period 3**. 6 outer electrons → **group 6**.

Section A: Quick recall (7 marks – 1 mark each)

1 What is a vertical column in the periodic table called?

2 What is a horizontal row in the periodic table called?

3 Are most elements metals or non-metals?

4 Where in the periodic table are the non-metals found?

5 What type of ions do metals form?

6 What do elements in the same group have in common?

7 Phosphorus has the electronic structure 2,8,5. Which group is it in?

Section B: Exam questions

01 Element X has the electronic structure 2,8,2.

01.1 Which group is element X in?

[1 mark]

01.2 Which period is element X in?

[1 mark]

01.3 Is element X a metal or a non-metal? Give a reason.

[2 marks]

02 Explain why sodium and potassium have similar chemical properties.

[2 marks]

03 Give two physical properties of metals that are different from non-metals.

[2 marks]

04 Silicon is in group 4 and period 3. Give the electronic structure of silicon and explain how you worked it out.

[3 marks]

Mark Scheme – Organisation of the Periodic Table

(1) = one mark. / separates acceptable alternatives. Words in brackets are not needed for the mark.

Question	Marking guidance	Comments	Mark
A	**1** a group **2** a period **3** metals **4** towards the right and top **5** positive ions **6** the same number of outer shell electrons / similar chemical properties **7** group 5	1 mark each	7
01.1	group 2 (1)		1
01.2	period 3 (1)		1
01.3	metal (1) it is in group 2 / on the left of the periodic table / it loses electrons to form positive ions (1)		2
02	they are both in group 1 (1) they both have one electron in their outer shell (1)		2
03	any two from: (1) each • conduct electricity / heat • shiny • malleable • high melting / boiling points		2
04	2,8,4 (1) period 3 means 3 shells (1) group 4 means 4 electrons in the outer shell (1)		3

Total: 18 marks